EBN111 Klastoets/*Class Test*#1: 2 Maart/*March* 2012

Naam/*Name*:	MEMORANDUM
Studente/*Student* nr:	
Graad/*Degree*:	B. Ing/*Eng*( )

Q1	Q2	Q3	Q4
t = 3 sec. ✓	t = 6.7 uur ✓✓	P_{10A} = −800 W ✓ $P_{2\Omega}$ = 2888 W ✓	V_{ab} = 5 V ✓✓

Q5	Q6	Q7	Q8
I_1 = −2.5 A ✓✓	V_2 = 5.71 V ✓✓	a = 19 b = −4 ✓✓ c = −6 d = 11	V_1 = 0.2 V ✓✓

Finale Punt/*Final Mark*: /15

(1) Indien die stroom in 'n geleier 2mA is, hoe lank sal dit neem voordat 6mC deur die geleier beweeg het/*If the current in a conductor is 2mA, how much time is required for 6mC to pass through the conductor?* [1]

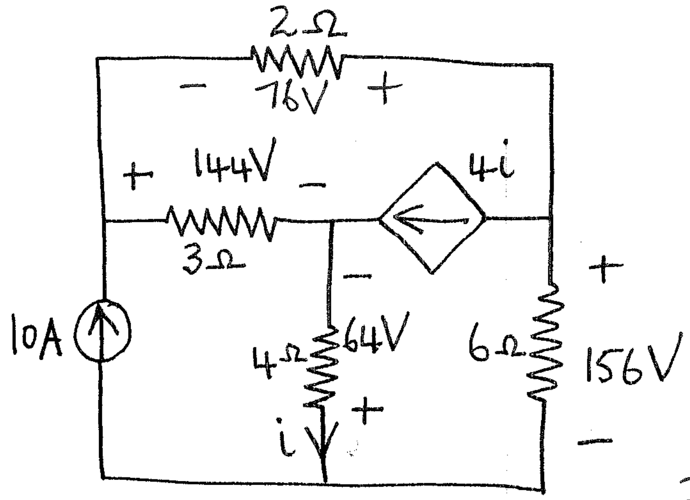
$$i = \frac{\Delta Q}{\Delta t} \Rightarrow \Delta t = \frac{\Delta Q}{i} = \frac{6mC}{2mA} = 3\text{ sec.} \rightarrow$$

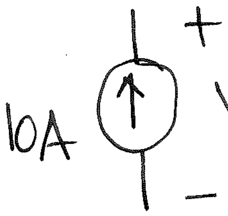
(2) Teen 60 sent per kWh, hoeveel ure kan jy 'n 250W TV-stel gebruik vir 'n totale koste van een rand/*At 60 cent per kWh, how many hours can you watch a 250W TV for one rand?* [2]

$\frac{1}{0.6} = 1.67$ kWh kan gekoop word met een rand

$$\frac{1.67}{0.25} = 6.7\text{ uur} \rightarrow$$

(3) Bereken die drywings ge-assosieer met die 10A bron en die 2Ω weerstand/*Calculate the power associated with the 10A source and the 2Ω resistor.* [2]





$V_{10A} = 144 - 64$

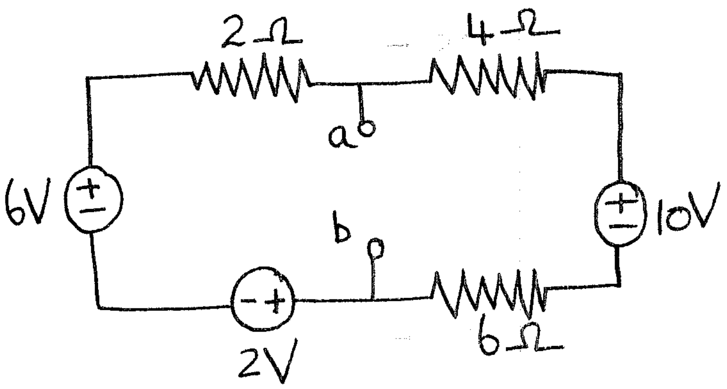
$= 80V$

$P_{10A} = -10 \times 80 = \underline{-800W}$

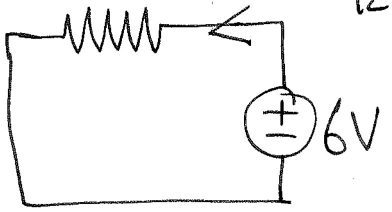
$3i + 10 = I_{2\Omega}$

$I_{2\Omega} = 3\left(-\frac{64}{4}\right) + 10 = -38A$ $\quad P_{2\Omega} = (+76) \times (-38) = \underline{2888W}$

(4) Bereken/*Calculate* V_{ab}. [2]



$=$

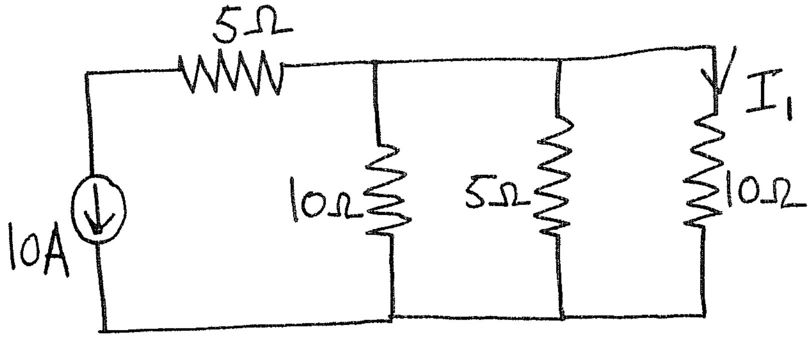


$i = \frac{6}{12} = 0.5A$

$-V_{ab} - (4+6) \times 0.5 + 10 = 0$

$V_{ab} = -5 + 10 = \underline{5V}$

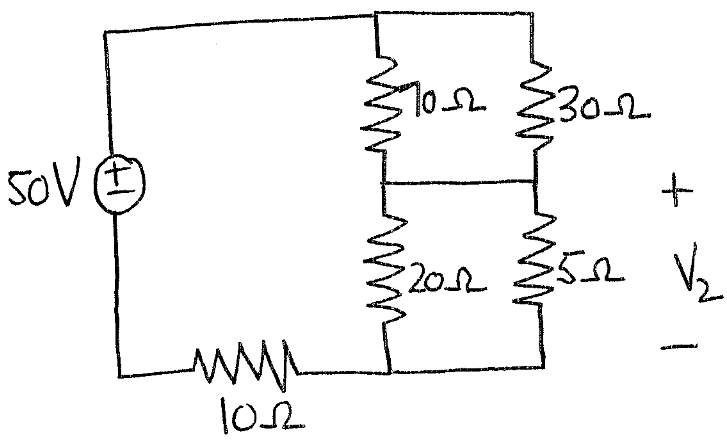
(5) Bereken/*Calculate* I_1. [2]



$$10\|5 = \frac{50}{15} = 3.33\,\Omega$$

$$I_1 = -10 \times \frac{3.33}{10+3.33} = -2.5\,A$$

(6) Bereken/*Calculate* V_2. [2]



$$30\|70 = 21\,\Omega$$

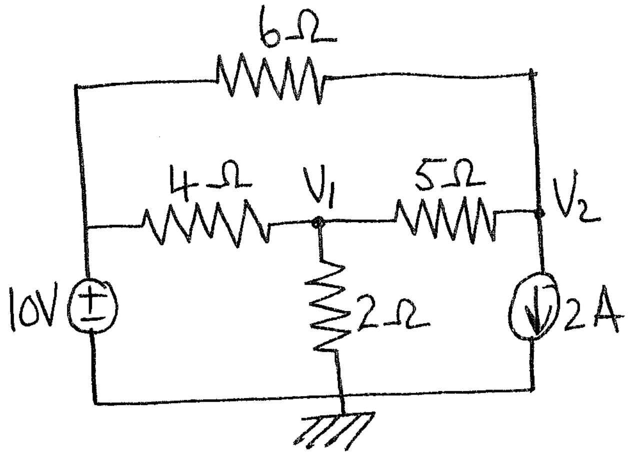
$$20\|5 = 4\,\Omega$$

$$V_2 = \frac{4}{21+4+10} \times 50 = 5.71\,V$$

(7) Gebruik-node-analise om twee vergelykings van die volgende vorm op te stel/*Use nodal analysis to set up two equations of the following form*:

$$aV_1 + bV_2 = 50$$
$$cV_1 + dV_2 = -10$$

[2]



KCL at node 1:

$$\frac{V_1 - 10}{4} + \frac{V_1}{2} + \frac{V_1 - V_2}{5} = 0$$

x20: $5V_1 - 50 + 10V_1 + 4V_1 - 4V_2 = 0$

$$19V_1 - 4V_2 = 50$$

KCL at node 2:

$$2 + \frac{V_2 - V_1}{5} + \frac{V_2 - 10}{6} = 0$$

x30: $60 + 6V_2 - 6V_1 + 5V_2 - 50 = 0$

$$-6V_1 + 11V_2 = -10$$

(8) Bepaal/*Determine* V_1: $\begin{bmatrix} 4 & -3 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$ [2]

$$V_1 = \frac{\begin{vmatrix} 2 & -3 \\ 0 & 2 \end{vmatrix}}{\begin{vmatrix} 4 & -3 \\ 4 & 2 \end{vmatrix}} = \frac{4 - 0}{8 - (-12)} = \frac{4}{20} = 0.2V$$